

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		2.0 [kg m/s]		1		1		
		(ii)		Substitution: $\frac{2.0 \text{ ecf}}{0.160}$ (1) = 12.5 [m/s] (1)	1	1		2	2	
		(iii)		0.0005 or 0.5×10^{-3} [s]		1		1		
		(iv)		Substitution: $\frac{2.0}{0.5 \times 10^{-3} \text{ ecf}}$ (1) = [-] 4 000 [N] (1) Note- Award 1 mark for answer 4 where n is not equal to 3	1	1		2	2	
	(b)	(i)		equal to (1) no (1)	2			2		
		(ii)		Straight diagonal line from (2.5, 0) → (3.0, 2.0) (1) Straight horizontal line from (3.0, 2.0) → (5.0, 2.0) (1) Award 1 mark only for diagonal line starting from (2.5,0) to any point for a momentum of 2 followed by a horizontal line			2	2		
				Question 1 total	4	4	2	10	4	0